

Formal verification of infinite state systems with **unbounded** agents

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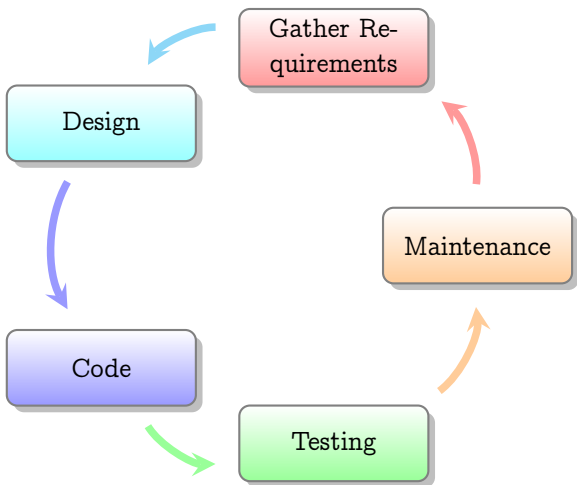
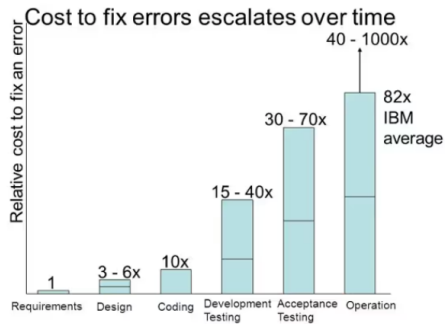


Figure: Traditional Software Development Life Cycle

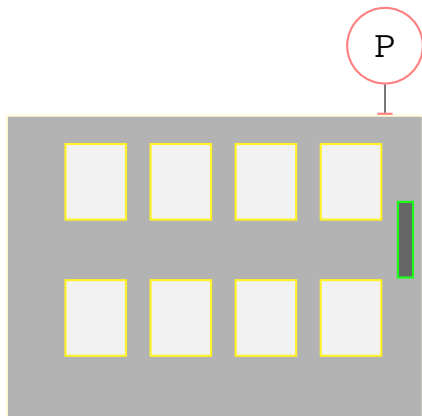
Error Costs



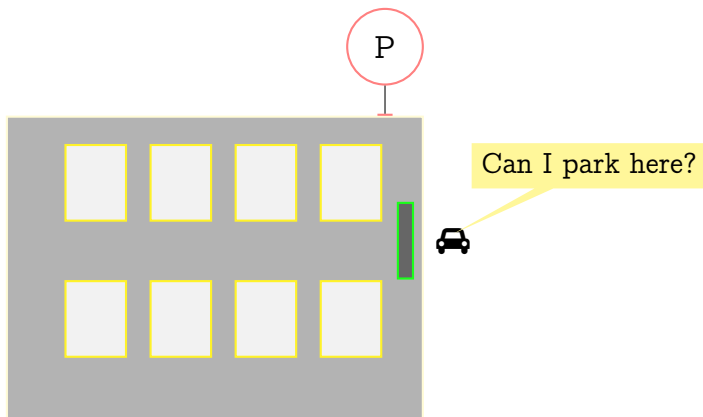
J. Stecklein, "Error cost escalation through the project life cycle, 2010

Running example: Unbounded Client-Server System (UCS)

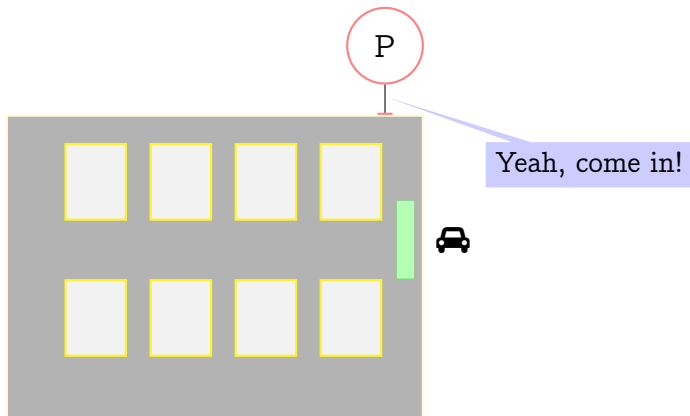
An empty Parking Lot waits for vehicles



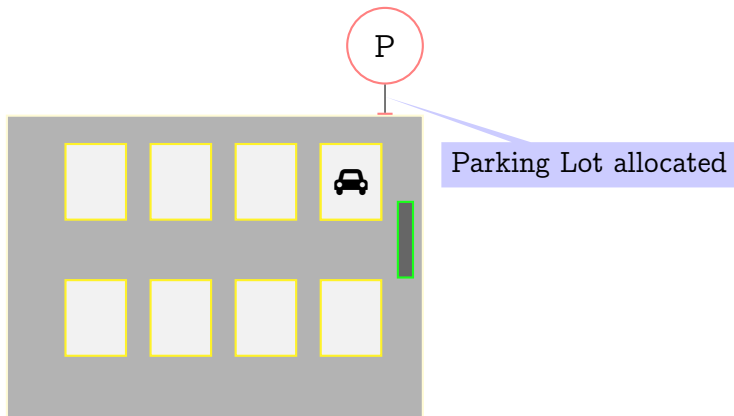
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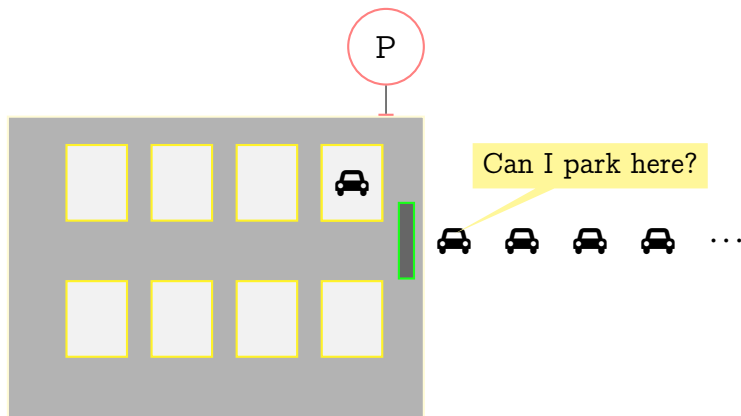


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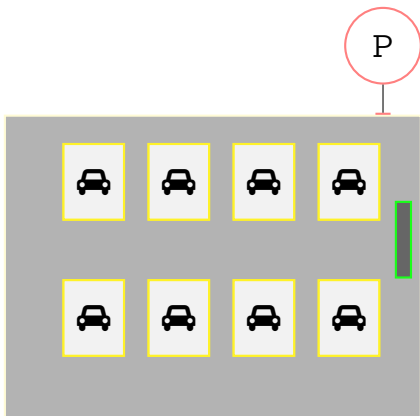
Running example: Unbounded Client-Server System (UCS)

Unbounded number of parking requests

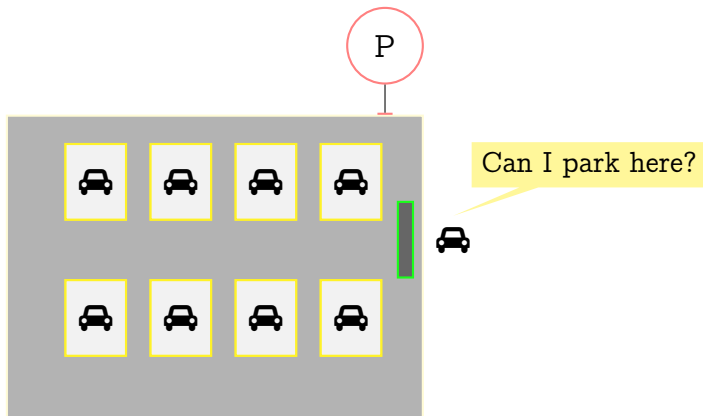


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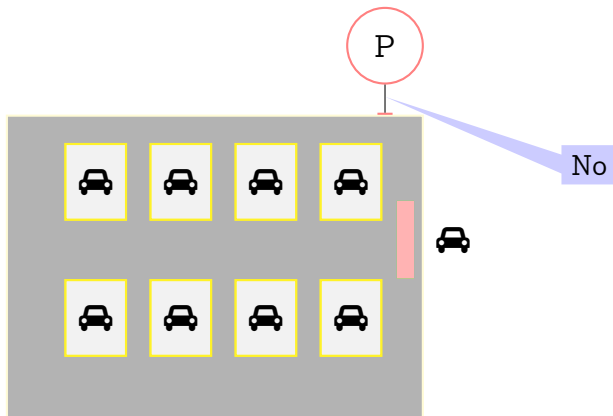
No parking space



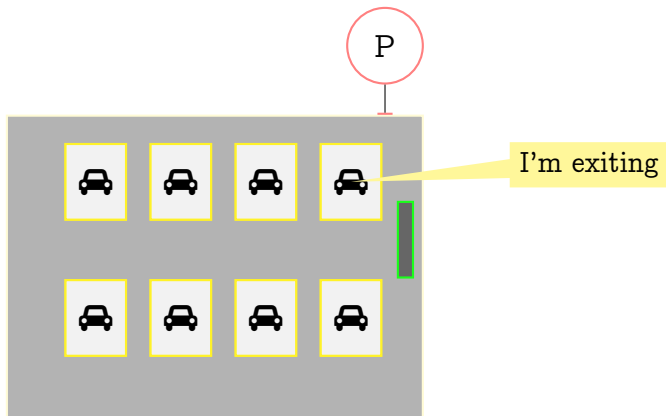
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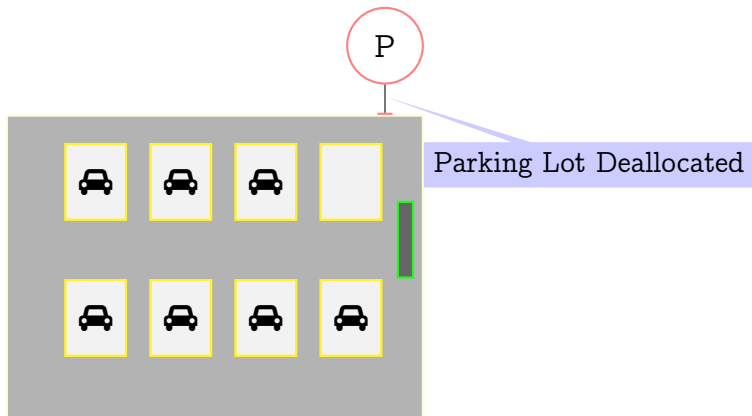
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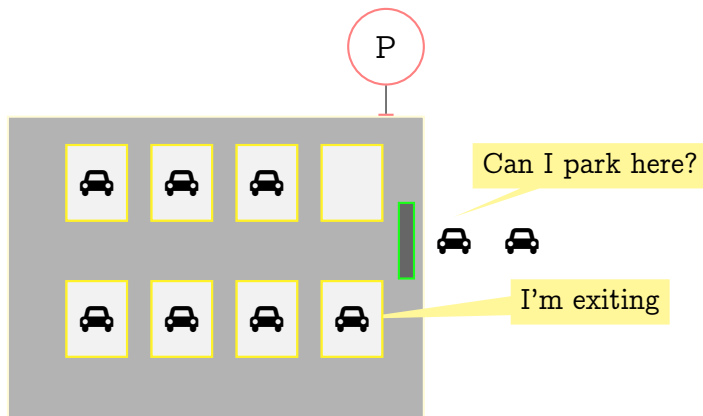


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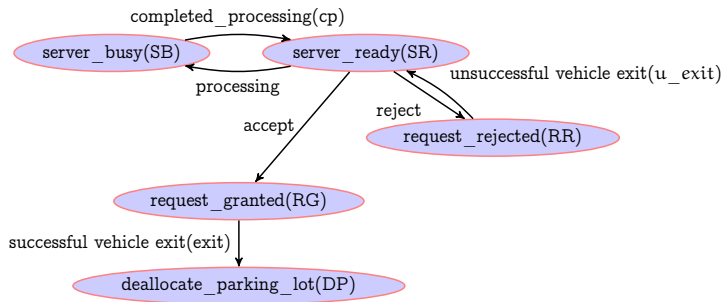


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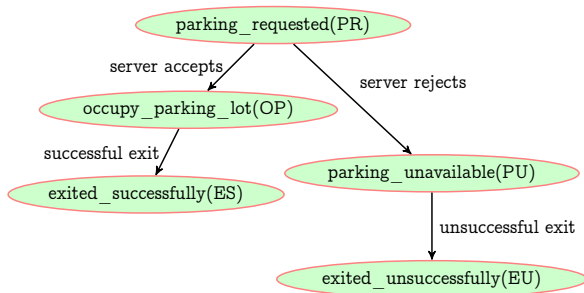
- single server-multiple client system
- **unbounded** clients of the same type



Case Study Parking System: Server behaviour



Case Study Parking System: Client behaviour



Research Questions

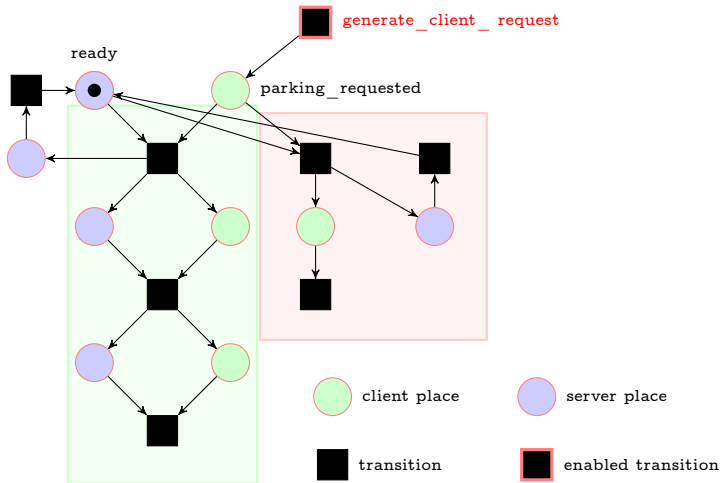
- 1 How do we formally verify UCS?
- 2 How do we formally model UCS?
- 3 How do we express properties of client-server systems naturally?
 - 1 Do existing logics suffice?
 - 2 What additional properties are interesting in the context of unbounded client-server systems?

Research Questions

- 1 How do we formally verify UCS?
- 2 How do we formally model UCS?
- 3 How do we express properties of client-server systems naturally?
 - 1 Do existing logics suffice?
 - 2 What additional properties are interesting in the context of unbounded client-server systems?
- 4 Are there formal verification tools for UCS? Can we add to the repertoire?
 - 1 How are existing tools different, unique? Is another tool necessary?

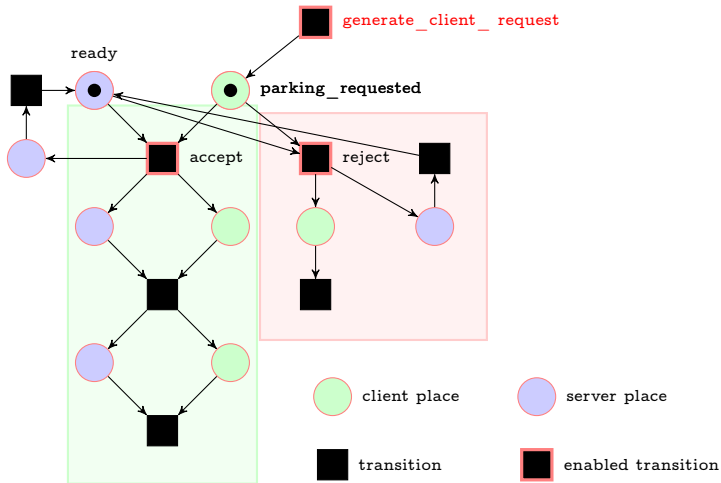
Challenge 1: Modelling UCS using Petri Nets

Initially the server is
ready for client requests



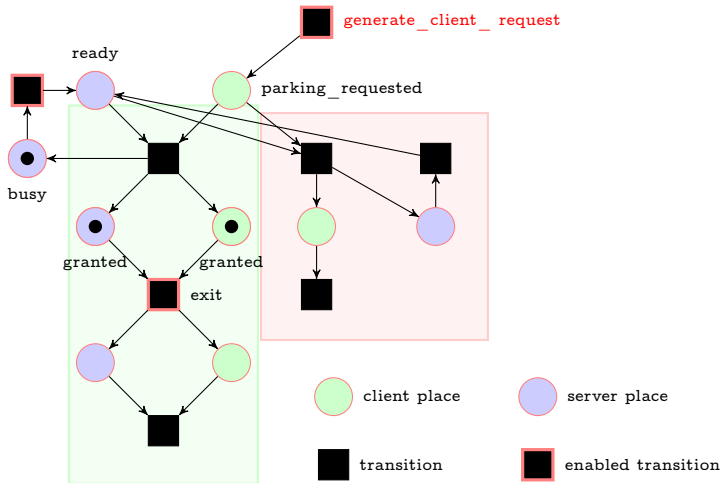
Challenge 1: Modelling UCS using Petri Nets

Client parking request is received



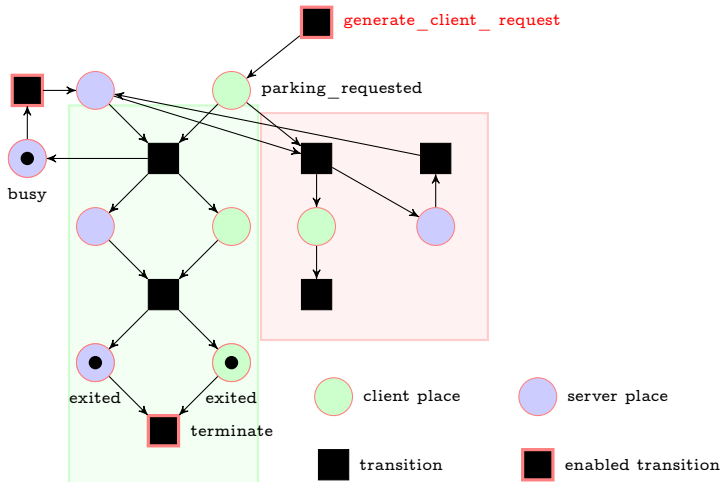
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Server accepts the Client
parking request



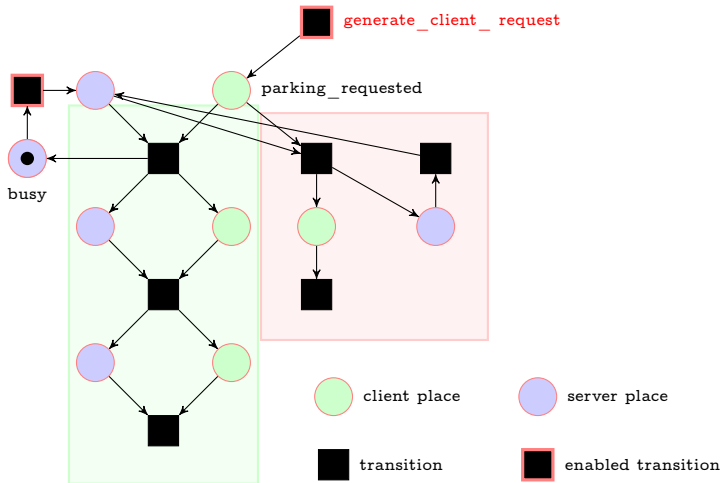
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Client exits the parking
space



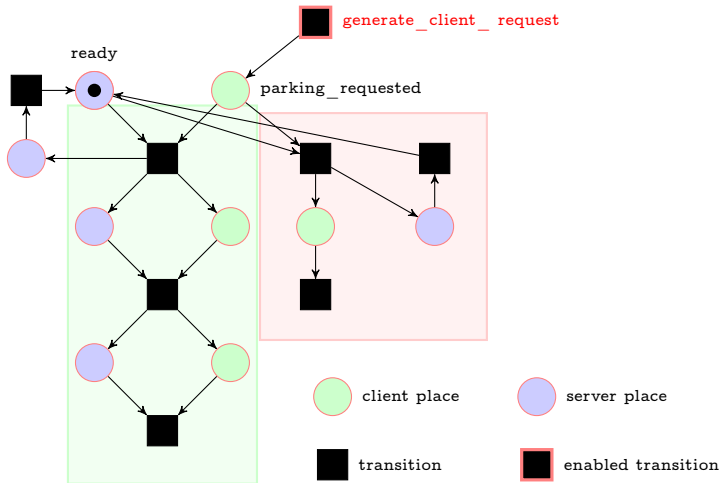
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Client is terminated



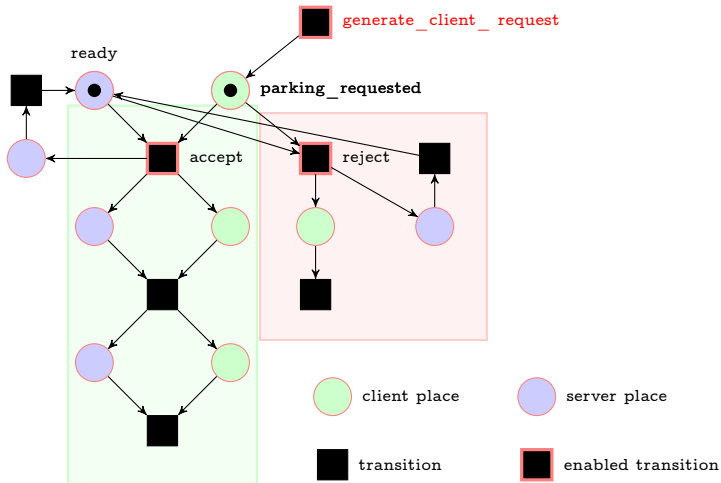
Challenge 1: Modelling UCS using Petri Nets

Server is ready for more requests



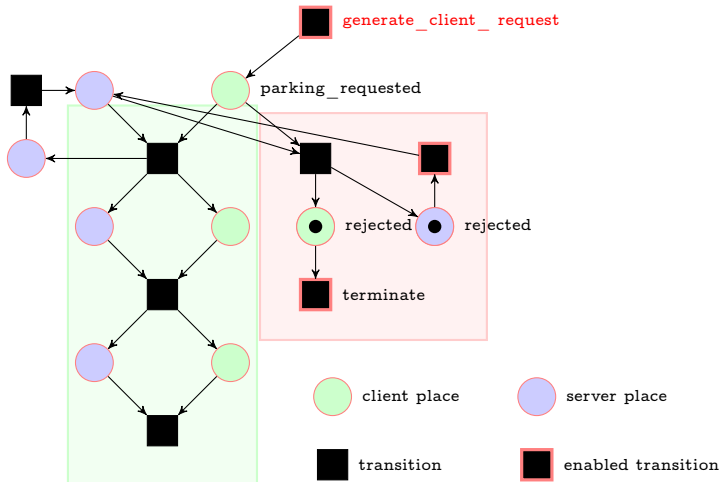
Challenge 1: Modelling UCS using Petri Nets

A new Client parking
request is received



Challenge 1: Modelling UCS using Petri Nets

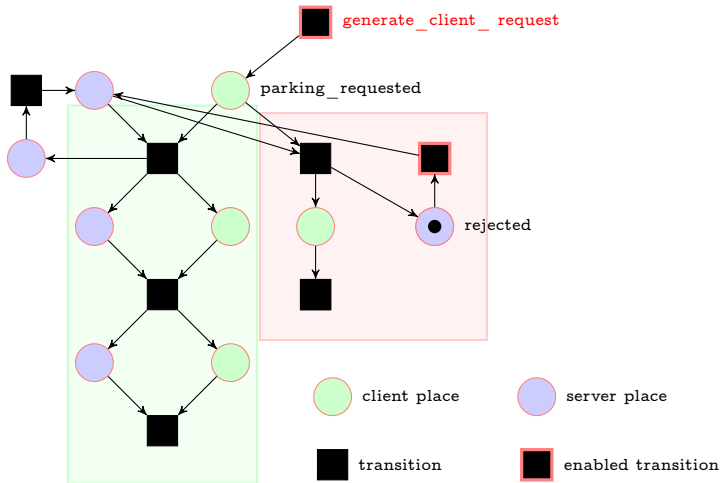
Server rejects the Client
parking request



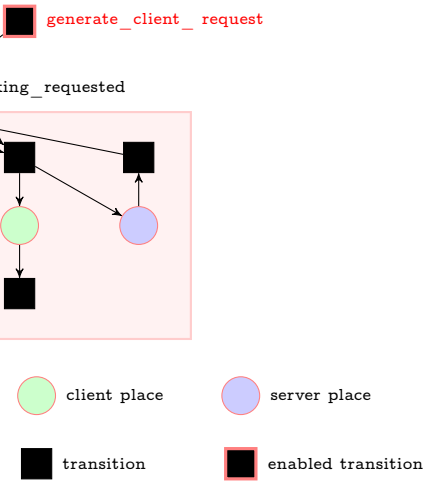
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Client is terminated

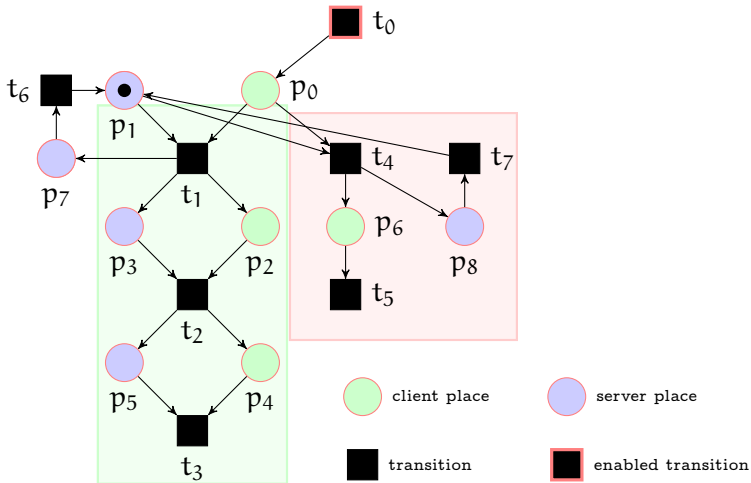
unsuccessfully



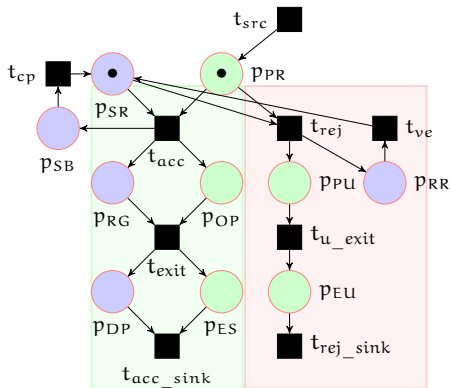
using Petri Nets



Challenge 1: Modelling UCS using Petri Nets



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Finite sets of atomic client propositions P_c and server propositions P_s :

$$P_c = \{\text{parking_requested}(PR), \text{occupy_parking_lot}(OP), \text{parking_unavailable}(PU), \text{exit_successfully}(ES), \text{exited_unsuccessfully}(EU)\}$$

$$P_s = \{\text{server_ready}(SR), \text{server_busy}(SB), \text{request_granted}(RG), \text{request_rejected}(RR), \text{deallocate_parking_lot}(DP)\}$$

Question: What properties of the system are we interested in?

Properties

- 1 Is the number of clients occupying parking lots = number of requests granted always?

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$$G(\#p_{PR} + \#p_{OP} + \#p_{ES} = 1).$$

- 3 Is the number of rejected requests greater than the number of accepted requests at some point?

$$F(\#p_{EU} > \#p_{ES}).$$

Linear Temporal Logic with Integer Arithmetic LTL_{LIA}

In LTL_{LIA} , there are two types of atomic formulas:

1 P_s - describing basic server properties, which are propositional constants and can be used to describe the transitions of the net

2 $\beta \in \Delta$, the set of client formulas

$$\alpha, \hat{\alpha} ::= \#p \mid c \mid (c * \#p) \mid (\#p * c) \mid (\alpha + \hat{\alpha}) \mid (\alpha - \hat{\alpha})$$

$$\beta ::= (\alpha < \hat{\alpha}) \mid (\alpha > \hat{\alpha}) \mid (\alpha \leq \hat{\alpha}) \mid (\alpha \geq \hat{\alpha}) \mid (\alpha = \hat{\alpha})$$

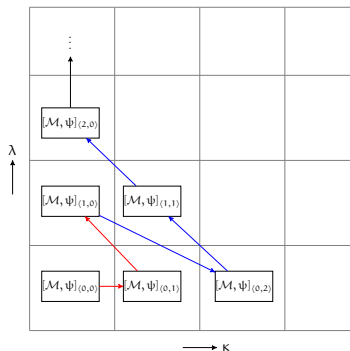
where $p \in P_c$, the set of client propositions and $c \in \mathbb{Z}^+$.

The set of server formulas Ψ are defined as follows:

$$\psi ::= q \in P_s \mid \beta \in \Delta \mid \neg\psi \mid \psi \vee \psi' \mid \psi \wedge \psi' \mid X\psi \mid F\psi \mid G\psi \mid \psi U \psi'$$

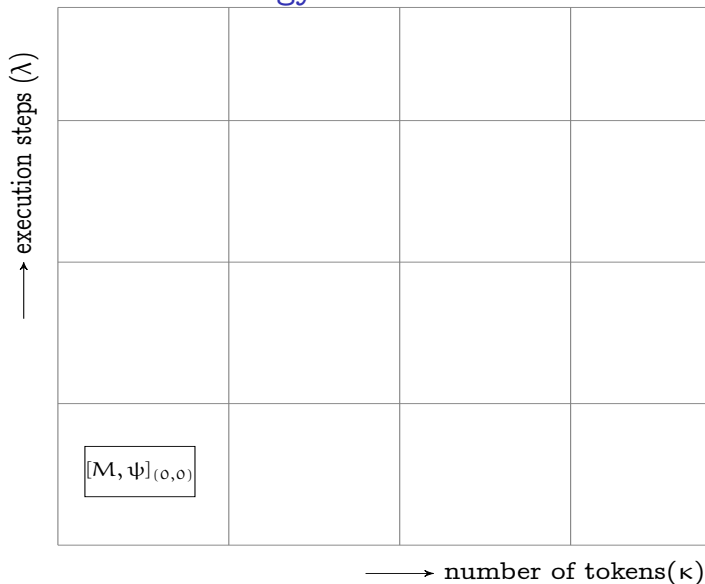
where $\psi, \psi' \in \Psi$. Modalities X , F , G and U are the usual modal operators: Next, Eventually, Globally and Until, respectively.

BMC vs 2D-BMC



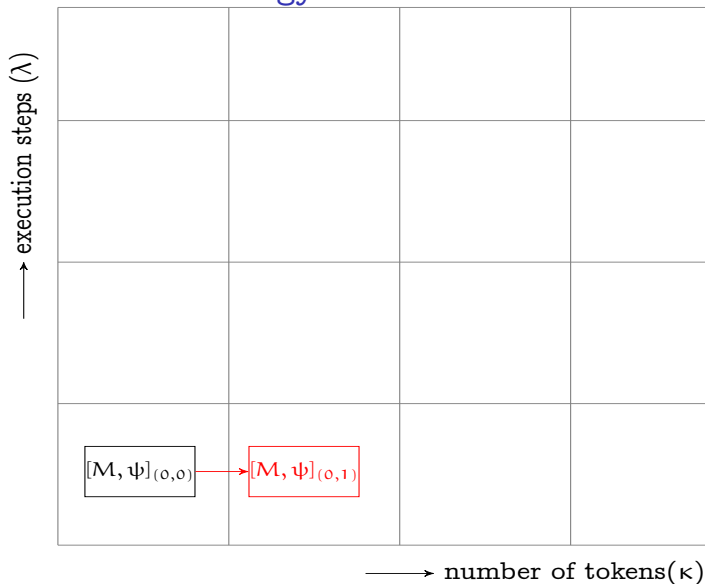
- In BMC, “We ask whether the system M has any counterexample of length k to (property) ψ . This bounded problem is encoded into SAT.” - A.Biere
- In 2D-BMC, we ask whether the system M has any counterexample of length λ and number of clients κ to ψ and encode this bounded problem into SMT.

The 2D-BMC strategy



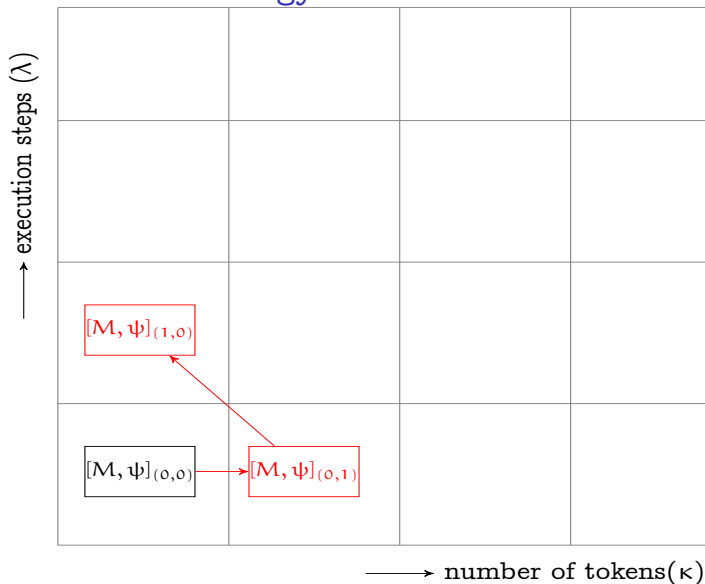
execution steps + number of clients = bound

The 2D-BMC strategy



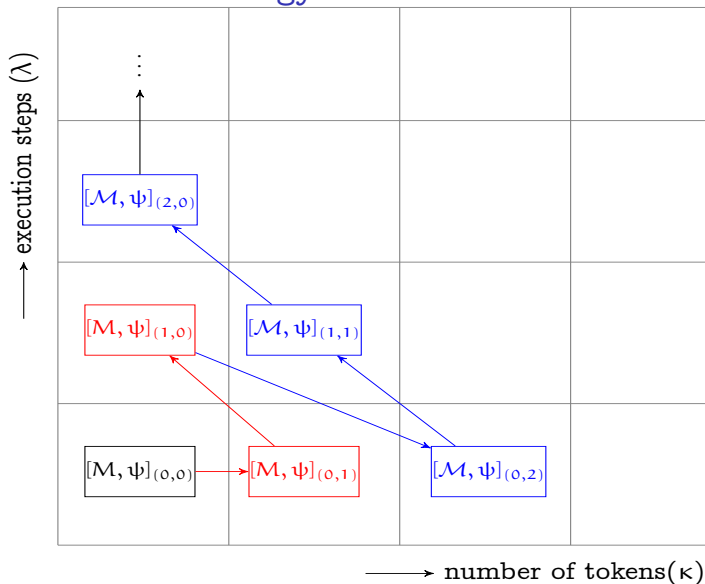
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Tool Architecture

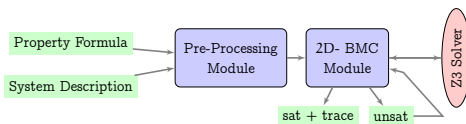


Figure: DCModelChecker architecture

Experimental Results

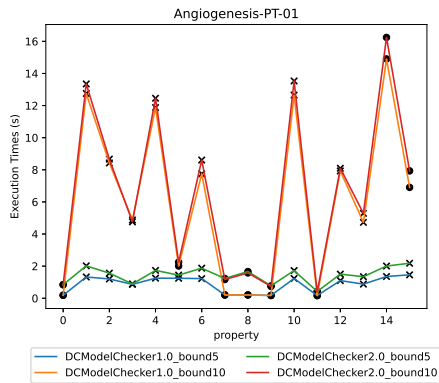


Figure: Verification of FireabilityCardinality Properties with bound 5 and 10

Verified against 16 Benchmarks from Model Checking Contest 2025

Experimental Results

Model	Execution Times(ms)			
	DCModelChecker 1.0	DCModelChecker 2.0	ITS-Tools	Tapaal
Parity	0.010	0.40	0.933	0.02
PGCD	0.019	0.98	1.487	0.07
Process	0.033	0.85	1.201	1e-05
CryptoMiner	0.036	0.64	0.957	7e-06
Murphy	0.031	0.83	1.161	8e-06

Table: Results of Comparative Experiments on Unbounded PNs

Contributions

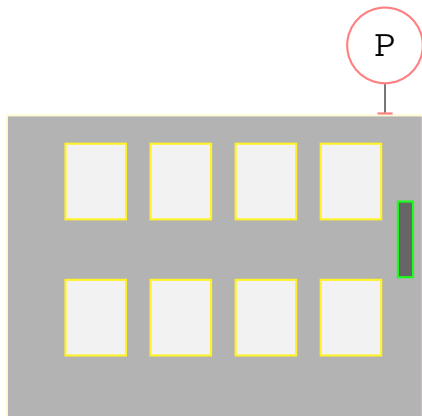
- Part 1: Verification of Unbounded Client-Server (UCS) using Petri Nets and Linear Temporal Logic with integer arithmetic using 2D-BMC

Challenge 2: Modelling an Unbounded Client-Server (UCS)

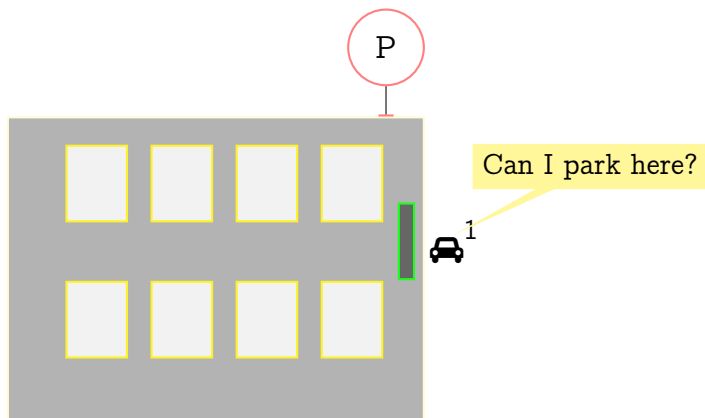
What if clients needed to be distinguished?

Running example

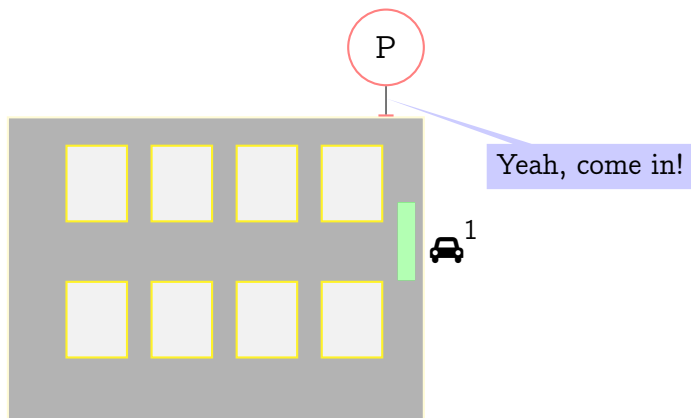
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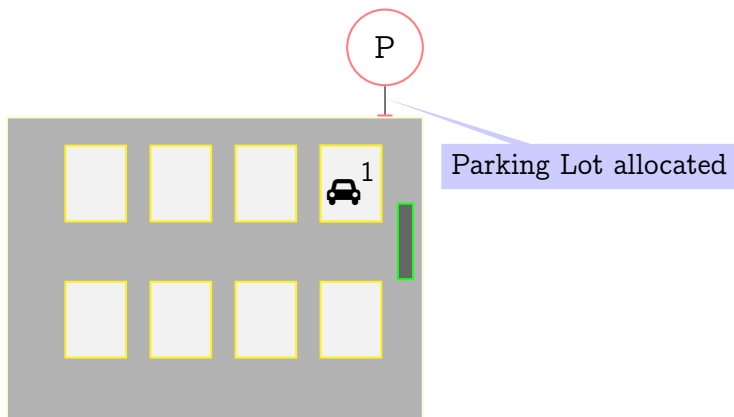
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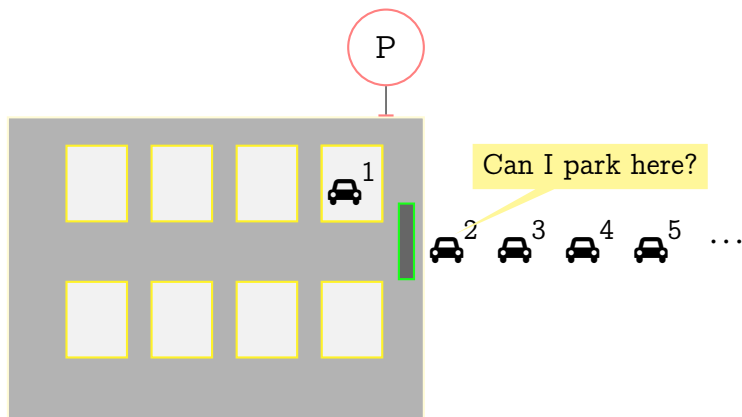


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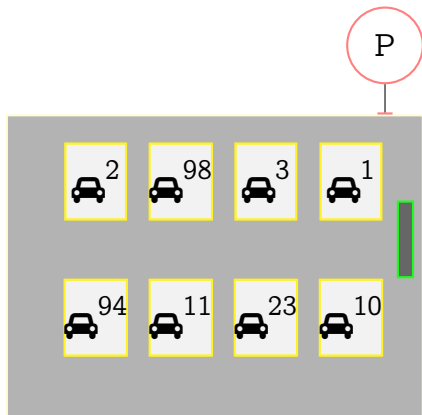
Running example

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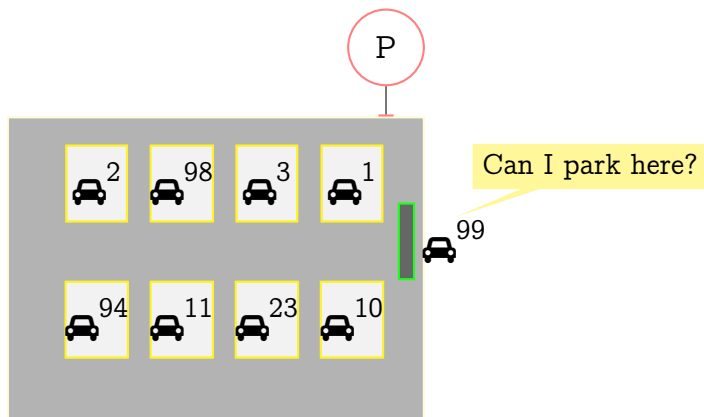


Running example

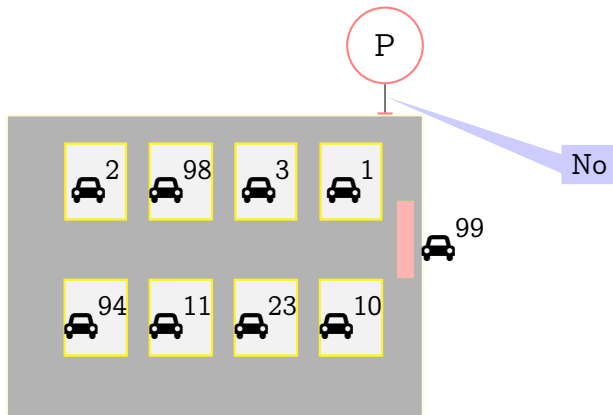
No parking space



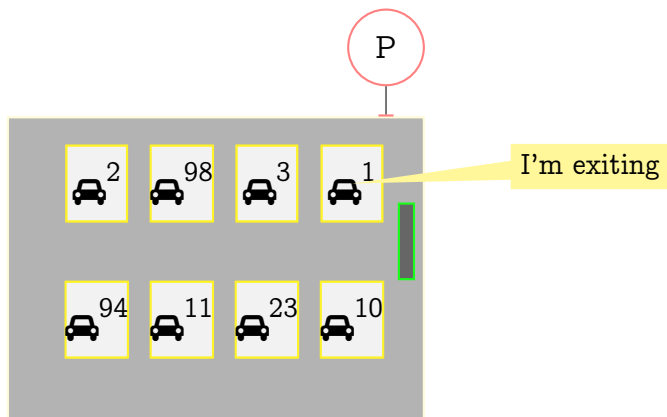
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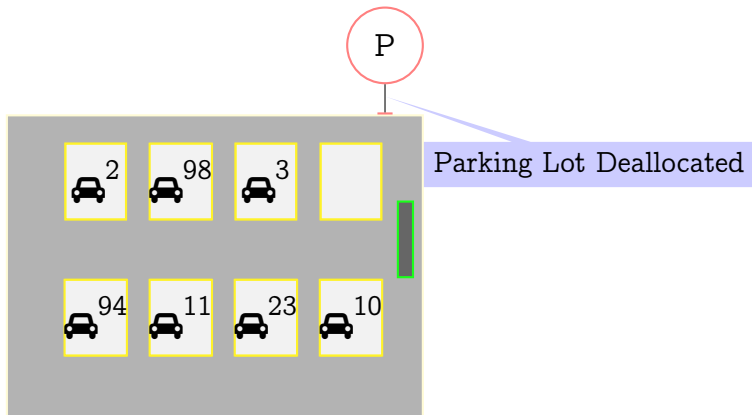
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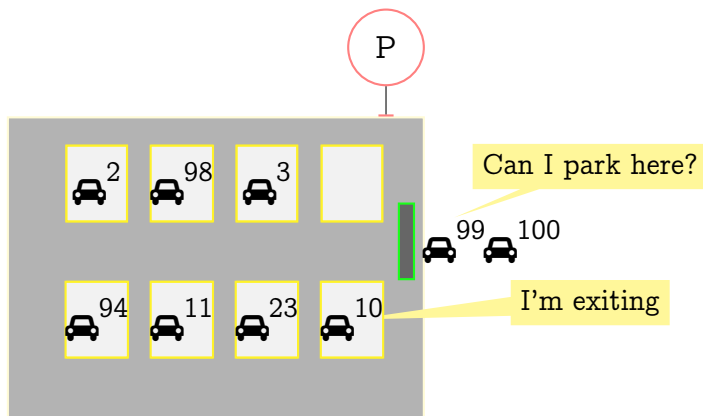


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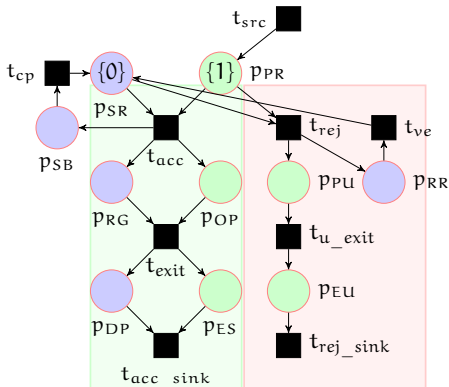


Running example

- single server-multiple client system
- unbounded distinguishable clients of the same type



Modelling an Unbounded Client-Server (UCS)



Finite sets of atomic client propositions P_c and server propositions P_s :

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$$G_s(\forall x) \left(\text{occupy_parking_lot}(x) \Rightarrow F_c(\text{exit_successfully}(x)) \right)$$

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The proposed logic: The Monodic* Logic FOTL₁

The set of client formulae Δ :

$$\Delta ::= p(x) \mid \neg\alpha \mid \alpha \vee \beta \mid \alpha \wedge \beta \mid X_c\alpha \mid F_c\alpha \mid G_c\alpha \mid \alpha U_c \beta$$

where $\alpha, \beta \in \Delta$ and $p \in P_c$, the set of atomic client propositions.

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The set of **server formulae** Ψ :

$$\Psi ::= q \mid \neg\psi \mid (\exists x)\alpha \mid (\forall x)\alpha \mid \psi_1 \vee \psi_2 \mid \psi_1 \wedge \psi_2 \mid \\ X_s\psi \mid F_s\psi \mid G_s\psi \mid \psi_1 U_s \psi_2$$

where $\psi, \psi_1, \psi_2 \in \Psi$ and $q \in P_s$, the set of atomic server propositions, $\alpha \in \Delta$.

The Monodic Logic FOTL_1 (contd)

$$\psi = (\exists x)(\exists y) \left((\text{PR}(x) \wedge \text{PR}(y)) \wedge \text{F}_c(\text{ES}(x) \wedge \text{ES}(y)) \right)$$

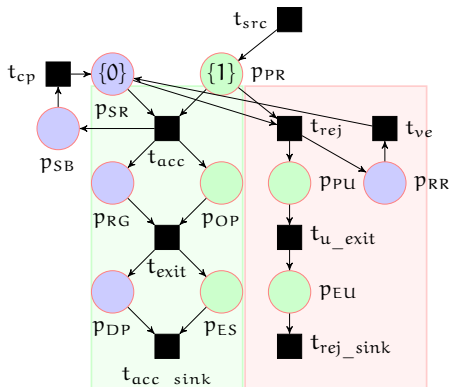
- Is $\psi \in \text{FOTL}_1$?

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- Is $\psi \in \text{FOTL}_1$?
- **Answer: No. not a well formed formula in FOTL_1**

Recall: our model

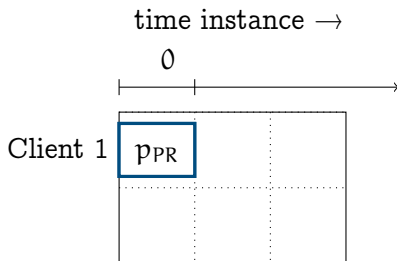


Atomic client propositions P_c
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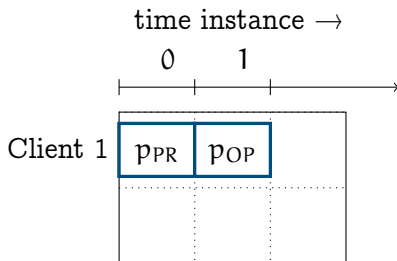
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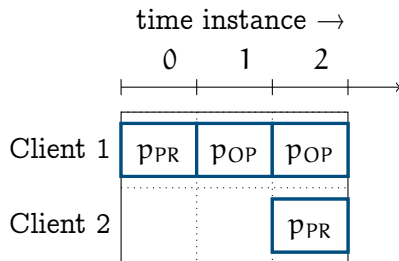
Snapshots of the system + live windows



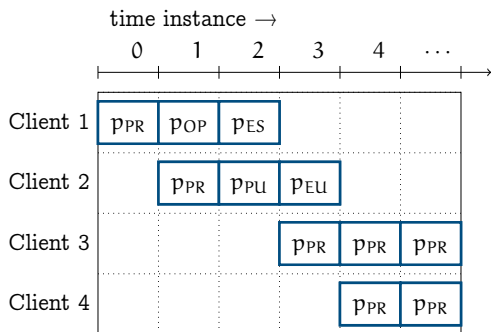
Snapshots of the system + live windows



Snapshots of the system + live windows



Snapshots of the system + live windows (contd)



- 4 distinguishable clients, with overlapping *live windows*, with the bound 5.
- client 1 is at p_{PR} at instance 0. i.e, $PR(x)$ is satisfied in client 1 at instance 0.
- For client 1, the *left boundary* is at instance 0 and its *right boundary* is at instance 2.

Formally speaking...

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- Let $CS = (Q, \Sigma, \delta, I, F)$ be a finite state machine describing the client behaviour .

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- Let CN be a countable set of client names .
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- Let $\mathcal{J} : CN \times \mathbb{N}_0 \rightarrow Q$ be a partial mapping describing the local state of each client $\alpha \in CN$ at an instance $i \in \mathbb{N}_0$.
 - For instance, $\mathcal{J}(\alpha, i) \in q$, means that the local state of each client α at instance i is state q , where $q \in Q$.

Formally speaking...

A model is a triple $M = (\nu, V, \xi)$ where

- ν gives the local behaviour of the server as follows:

$\nu = \nu_0 \nu_1 \nu_2 \dots$, where for all $0 \leq i$, $\nu_i \subseteq P_s$

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For every $0 \leq i$, V_i and V_{i+1} satisfy the following properties:

- 1 if $V_i \subseteq V_{i+1}$ then for every $a \in V_{i+1} - V_i$ such that $\exists(a, i+1) \in I$.
- 2 if $V_{i+1} \subseteq V_i$ then for every $a \in V_i - V_{i+1}$ such that $\exists(a, i) \in F$.

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- $\xi = \xi_0 \xi_1 \xi_2 \dots$, where for all $0 \leq i$, $\xi_i : V_i \rightarrow 2^{P_c}$ gives the properties satisfied by each live agent at i th instance.

Also, $L(\exists(a, i)) = \xi_i(a)$.

Semantics

$M, i \models (\exists x)\alpha$ iff $\exists a \in CN, a \in V_i$ and $M, [x \mapsto a], i \models_x \alpha$.

Semantics

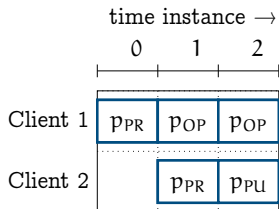
$M, i \models (\exists x)\alpha$ iff $\exists a \in CN, a \in V_i$ and $M, [x \mapsto a], i \models_x \alpha$.

$M, i \models^k (\exists x)\alpha$ iff $\exists a \in CN, a \in V_i$ and $M, [x \mapsto a], i \models_x^k \alpha$.

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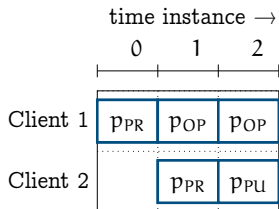


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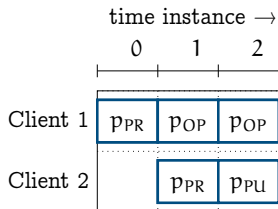
Given $\alpha = \text{OP}(x) \vee \text{PR}(x)$ and $\text{CN} = \{1, 2\}$.



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$M, i \models (\exists x)\alpha$ iff $\exists a \in CN, a \in V_i$ and $M, [x \mapsto a], i \models_x \alpha$.

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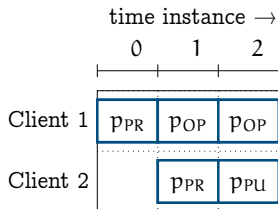


Given $\alpha = OP(x) \vee PR(x)$ and $CN = \{1, 2\}$.

- $M, 0 \models^k (\exists x)(OP(x) \vee PR(x))$.
- $M, 1 \models^k (\exists x)(OP(x) \vee PR(x))$.
- $M, 2 \models^k (\exists x)(OP(x) \vee PR(x))$.

Bounded Semantics

$M, i \models^k (\forall x)\alpha$ iff $\forall a \in \text{CN}$, if $a \in V_i$ then $M, [x \mapsto a], i \models_x^k \alpha$



Given $\alpha = \text{ES}(x)$ and $\text{CN} = \{1, 2\}$.

- $M, 0 \not\models^k (\exists x)(\text{ES}(x))$.
- $M, 1 \not\models^k (\exists x)(\text{ES}(x))$.
- $M, 2 \not\models^k (\exists x)(\text{ES}(x))$.

Bounded Semantics

$M, i \models^k F_s \psi$ iff $\exists j : i \leq j \leq k, M, j \models^k \psi$.

Given $\psi = (\exists x)ES(x)$.

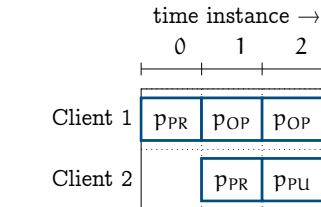
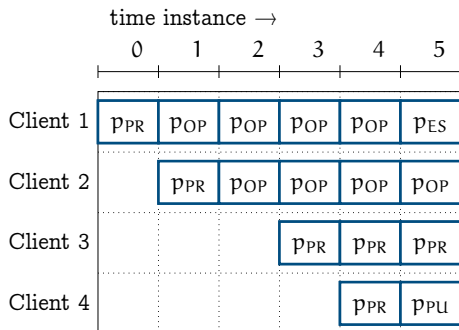


Figure: Snapshot with 2 clients

Figure: Snapshot with 4 clients

Bounded Semantics

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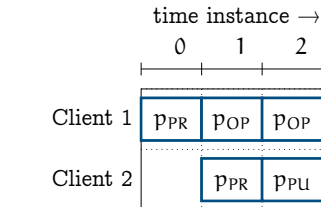
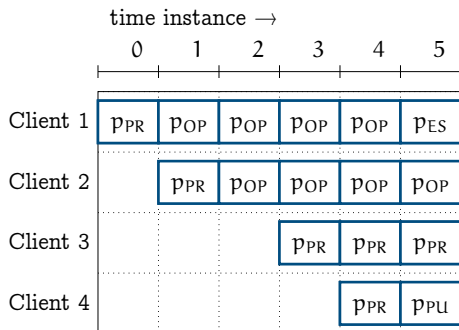


Figure: Snapshot with 2 clients

Figure: Snapshot with 4 clients

As shown above $M, 5 \models^k F_s (\exists x)ES(x)$.

Implementation

Verifying $G_s \exists x (F_c(p_0(x)))$

```
evname: cToken_4_1
evname: cToken_4_2
evname: cToken_4_3
evname: cToken_4_4
evname: cToken_4_5
evname: sToken_4_1
LOG: done with initializing empty token variables
LOG: sysObj.MAX_TOKENS_LIMIT before initial marking: 1
LOG: constructed initial Marking
LOG: constructed the complete Transition Function
LOG: nochange constraint added
LOG: UNSAT
LOG: UNSAT
LOG: UNSAT
LOG: UNSAT
LOG: -----run 5 -----
evname: cToken_5_0
evname: cToken_5_1
evname: cToken_5_2
evname: cToken_5_3
evname: cToken_5_4
evname: cToken_5_5
evname: sToken_5_1
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LOG: UNSAT
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```

Contributions

- Part 1: Verification of UCS using Petri Nets and LTL_{LIA}
- Part 2: Verification of UCS with distinguishable clients using ν -nets and properties specified in $FOTL_1$

Challenge 3: Modelling an Unbounded Client-Server (UCS)

What if clients had an internal structure?

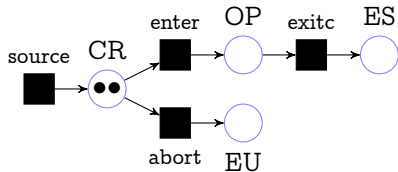


Figure: Petri net N_3 depicting client behaviour

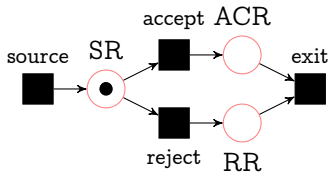


Figure: Petri net N_3 depicting server behaviour

Question: What properties of the system are we interested in?

- **Reachability:** Is $\mu_1 = \langle \text{SR}, \{\{\text{client1CR}, \text{client2CR}\}\} \rangle$ a reachable marking?
- **Coverability:** Given $\mu_1 = \langle \text{SR}, \{\{\text{client1CR}, \text{client20P}\}\} \rangle$ and $\mu_2 = \langle \text{SR}, \{\{\text{client10P}, \text{client2CR}, \text{client30P}\}\} \rangle$.

Then, is $\mu_1 \preceq \mu_2$?

Modelling UCS via Elementary Object Systems

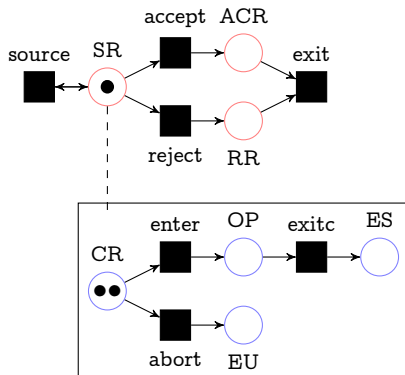
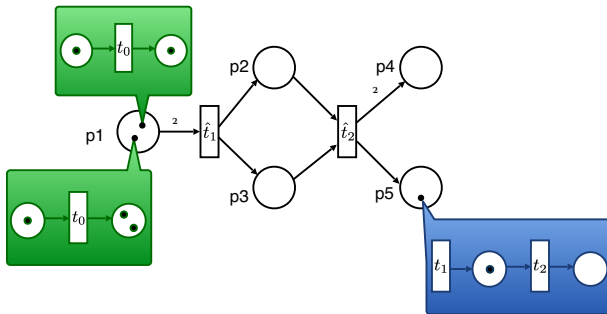
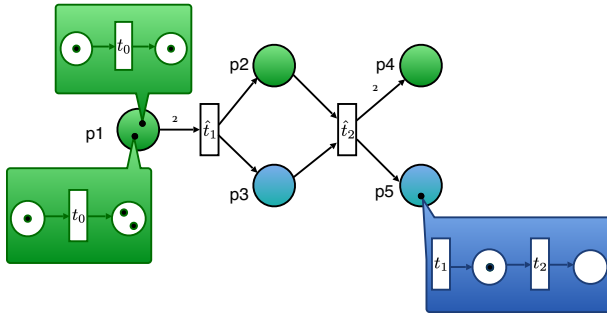


Figure: EOS modeling the single server (unbounded) multiple client system

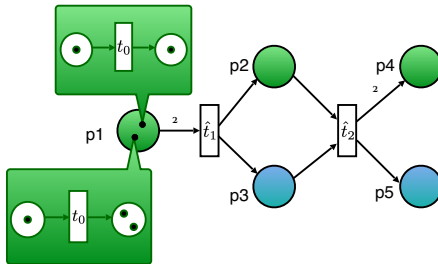
EOS – nested markings



EOS – typed places



EOS – object autonomous events



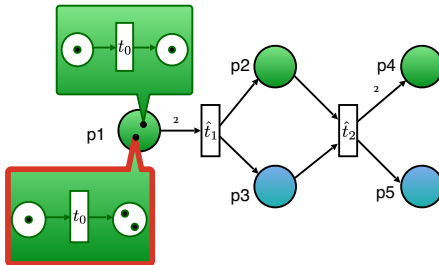
EVENTS

$$e_1 = (id_{p_1}, t_0)$$

$$e_2 = (\hat{t}_1, \emptyset)$$

$$e_3 = (\hat{t}_2, t_0 t_1 t_1)$$

EOS – object autonomous events



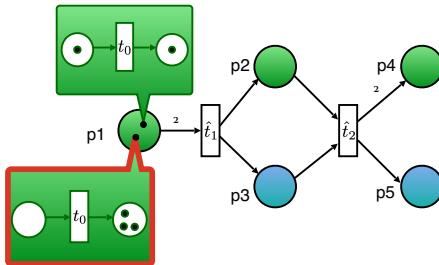
EVENTS

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$$e_2 = (\hat{t}_1, \emptyset)$$

$$e_3 = (\hat{t}_2, t_0 \hat{t}_1 \hat{t}_1)$$

EOS – object autonomous events



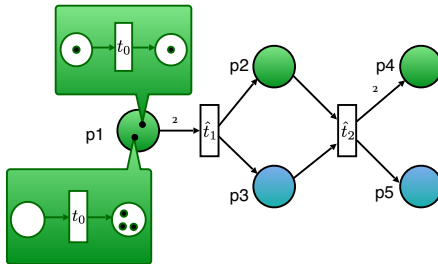
EVENTS

$$e_1 = (id_{p_1}, t_0)$$

$$e_2 = (\hat{t}_1, \emptyset)$$

$$e_3 = (\hat{t}_2, t_0 t_1 t_1)$$

EOS – system autonomous events



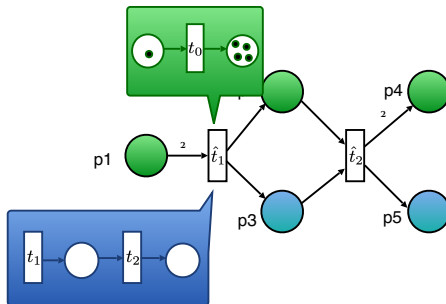
EVENTS

$$e_1 = (id_{p_1}, t_0)$$

$$e_2 = (\hat{t}_1, \emptyset)$$

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EOS – system autonomous events



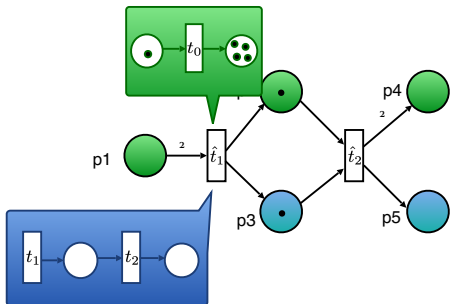
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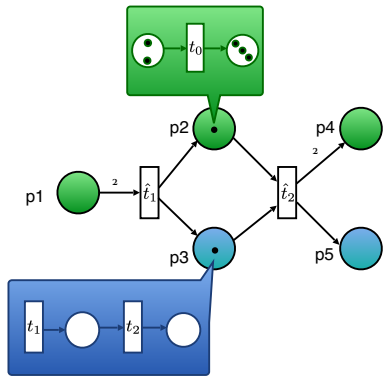
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EOS – system autonomous events



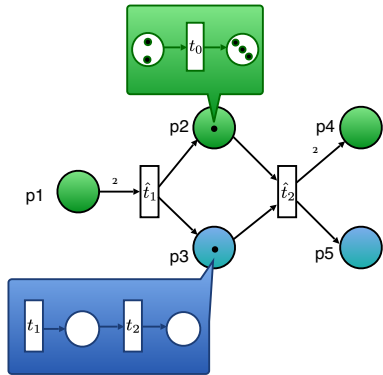
EVENTS

$$e_1 = (id_{p_1}, t_0)$$

$$e_2 = (\hat{t}_1, \emptyset)$$

$$e_3 = (\hat{t}_2, t_0 t_1 t_1)$$

EOS – synchronization events



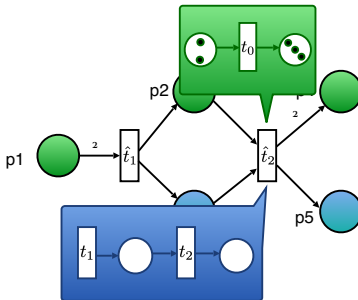
EVENTS

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$$e_2 = (\hat{t}_1, \emptyset)$$

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EOS – synchronization events



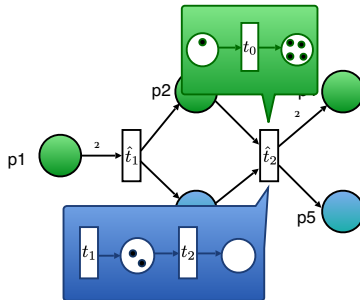
EVENTS

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$$e_2 = (\hat{t}_1, \emptyset)$$

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EOS – synchronization events



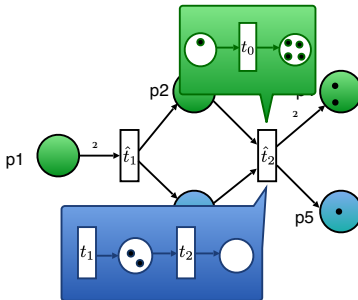
EVENTS

$$e_1 = (id_{p_1}, t_0)$$

$$e_2 = (\hat{t}_1, \emptyset)$$

$$e_3 = (\hat{t}_2, t_0 t_1 t_1)$$

EOS – synchronization events



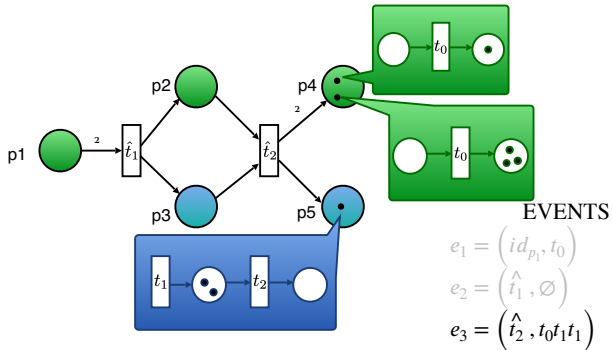
EVENTS

$$e_1 = (id_{p_1}, t_0)$$

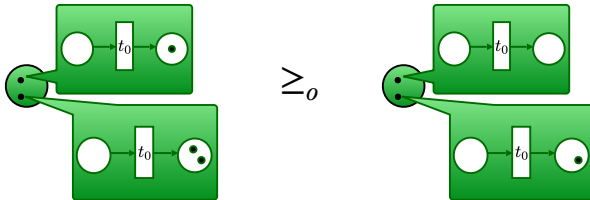
$$e_2 = (\hat{t}_1, \emptyset)$$

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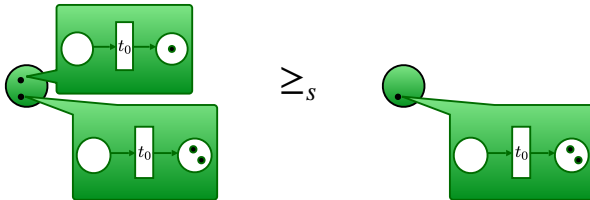
EOS – synchronization events



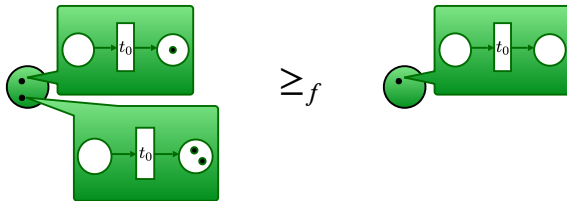
Object lossiness



System lossiness



Full lossiness

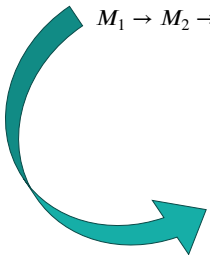




$(\preceq, \ell)$ -lossy runs

Perfect runs: only standard steps

$M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow \dots$



$(\preceq, \ell)$ -runs: at most $\ell \leq |\mathbb{N}|$ steps of type $\preceq$
 $M_1 \rightarrow M_2 \succcurlyeq M'_3 \rightarrow M'_4 \succcurlyeq M'_5 \succcurlyeq M'_6 \rightarrow M_7 \rightarrow \dots$

$(\preceq, \ell)$ -lossy problems



Is the system **robust** up to ℓ occurrences of $\preceq$?



$(\preceq, \ell)$ -lossy problems

Is the system **robust** up to ℓ occurrences of $\preceq$?

$(\preceq, \ell)$ -*deadlock freeness*

Input

An EOS E and an initial marking M .

Output

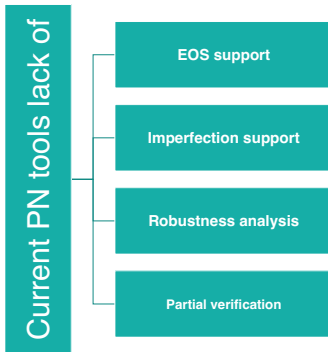
Is there a $(\preceq, \ell)$ -run from M to a marking where no event is enabled?

Decidability Results (PNSE'24)

	Problems	$\leq f$	$\leq o$	$\leq s$
Conservative EOS	0-reach	Undec [Buß14]	Undec [Buß14]	Undec [Buß14]
	cover	Dec [Buß14]	Undec (2CM)	Undec (2CM)
	f-reach/cover	Dec (Comp.)	Undec (Comp.)	Undec (Comp.)
	ω -reach/cover	Dec (Comp.)	Undec (Comp.)	Undec (Comp.)
EOS	0-reach	Undec [Buß14]	Undec [Buß14]	Undec [Buß14]
	cover	Undec [Buß14]	Undec (cEOS)	Undec (cEOS)
	f-reach/cover	Undec (Gadget)	Undec (cEOS)	Undec (Comp.)
	ω -reach/cover	Dec (WSTS)	Undec (cEOS)	Undec (Comp.)

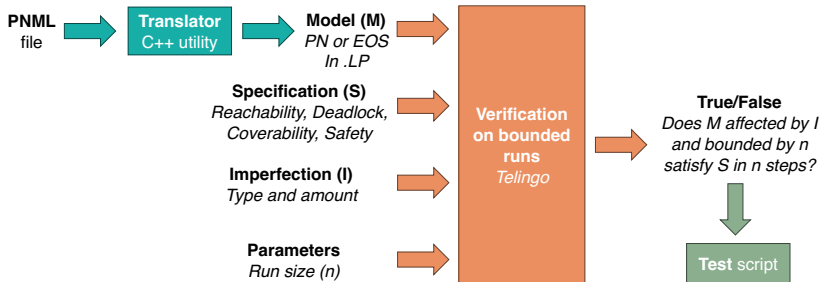
[Buß14] Köhler-Bußmeier, Michael. 'A Survey of Decidability Results for Elementary Object Systems'. 1 Jan. 2014 : 99 – 123.

Motivation





Prototype (CILC'24)



Why bounded verification?



Problems on general EOS	$\leq_f$	$\leq_o$	$\leq_s$
0-reach	U	U	U
0-cover	U	U	U
ℓ -reach/cover	U	U	U
ω -reach/cover	D	U	U

F. Di Cosmo, S. Mal, T. Prince, *Deciding Reachability and Coverability in Lossy EOS*, PNSE'24

Why Telingo?



Telingo is **declarative** and supports **temporal constraints**

- E.g., `:- &tel(>? (lossy >(>? lossy)` allows at most one lossy step
- The meaning of lossy is declared orthogonally to EOS specification

Telingo **returns finite runs**

- Perfectly matches bounded verification

Encoding of PNs and EOSs is **elegant in ASP**

- E.g., when compared to SMT – R. Phawade, T. Prince, S. Sheerazuddin et al., *Bounded Model Checking for Unbounded Client Server Systems*, Arxiv (2022)



Correctness and performances

Correct answers

- Checked on MCC benchmarks

Slow on PNs

- Compared with Tapaal

Prohibitively slow on EOSs

- Nesting exacerbates grounding

problem	lossiness	-imax =5 (s)	-imax =10 (s)	-imax =20 (s)	TAPAAL (s)
deadlock	none	UNSAT in 0.052	SAT in 0.622	SAT in 82.754	SAT in $5e-6$
deadlock	any	SAT in 0.009	SAT in 0.010	SAT in 0.009	NA
1-safeness	none	UNSAT in 0.010	UNSAT in 0.014	UNSAT in 0.027	UNSAT in 0
1-safeness	any	UNSAT in 0.013	UNSAT in 0.018	UNSAT in 0.032	NA

Table 1

Comparative Results with TAPAAL for the Eratosthenes-PT-010 PN from the MCC benchmarks [12].

Contributions

- Part 1: Verification of Unbounded Client-Server (UCS) using Petri Nets and Linear Temporal Logic with integer arithmetic
- Part 2: Verification of UCS with distinguishable clients using v-nets and properties specified in monodic First Order Logic
- Part 3: Verification of UCS where the clients have internal structure using Elementary Object Systems and properties specified in Linear Temporal Logic

List of publications:

- 1 Ramchandra Phawade, Tephilla Prince, S. Sheerazuddin: Bounded Model Checking for Unbounded Client Server Systems. Arxiv (2026), submitted to TopNOC
- 2 Ramchandra Phawade, Tephilla Prince, S. Sheerazuddin: Verification of Unbounded Client-Server Systems with Distinguishable Clients Arxiv (2026), submitted to TopNOC
- 3 Tephilla Prince: On Verifying Unbounded Client-Server Systems. Springer EUMAS 2023: 465-471
- 4 Francesco Di Cosmo, Tephilla Prince: Bounded Verification of Petri Nets and EOSs using Telingo: An Experience Report. CILC 2024

List of publications (ongoing):

- 5 Francesco Di Cosmo, Soumodev Mal, Tephilla Prince:
Deciding Reachability and Coverability in Lossy EOS.
PNSE@Petri Nets 2024: 74-95
- 6 Francesco Di Cosmo, Soumodev Mal, Tephilla Prince:
Decidability and Complexity of Reachability and
Coverability in Lossy EOS (Yet to be submitted to
Fundamenta Informatica)

Appendix

Bounded Semantics

$M, [x \mapsto a], i \models_x^k F_c \alpha$ iff

$\exists j : i \leq j \leq k, a \in V_j$ and $M, [x \mapsto a], j \models_x^k \alpha$.

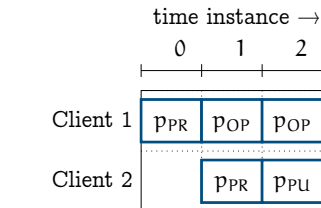
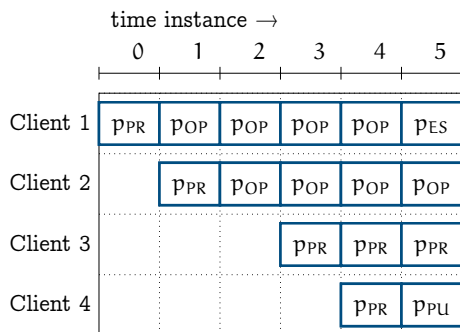


Figure: Snapshot with 2 clients

Figure: Snapshot with 4 clients

Given $\alpha = PR(x)$. The above formula is satisfiable for all clients in both figures.